

Hypothesis Testing



The logic of hypothesis testing

- 1 State the null and alternative hypotheses for testing whether a new fertilizer increases the mean crop yield μ , given a historical mean $\mu_0 = 50$ kg. Classify the test as one-tailed or two-tailed.
- 2 Explain the difference between a one-tailed and a two-tailed hypothesis test, giving an example scenario for each.
- 3 A test has $H_0 : \mu = 75$ vs $H_a : \mu \neq 75$, and the sample data leads to rejecting H_0 at the 5% significance level. State, without further calculation, whether this data would also lead to rejection at the 1% level, justifying your answer.

Hypothesis test for a population mean (known σ)

- 4 A machine fills bags with a claimed mean of 500g, with known population standard deviation $\sigma = 15$ g. A sample of $n = 36$ bags has mean $\bar{x} = 494$ g. Test at the 5% level whether the mean is less than 500g, using $z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$ and critical value -1.645 .
- 5 A population has known standard deviation $\sigma = 8$. A sample of $n = 64$ has mean $\bar{x} = 103$, against a claimed mean $\mu_0 = 100$. Test at the 1% level whether the mean exceeds 100, using critical value $z^* = 2.326$.

Two-tailed tests for a population mean

- 6 A standardized test historically has mean $\mu_0 = 100$ with $\sigma = 15$. A sample of $n = 25$ students taking a new prep course has mean $\bar{x} = 107$. Test at the 5% level (two-tailed) whether the course changes the mean score, using critical values ± 1.96 .
- 7 Using $z \approx 2.33$ from the previous problem, find the two-tailed p -value, given $\Phi(2.33) \approx 0.9901$, and confirm the conclusion.
- 8 A sample of $n = 50$ has mean $\bar{x} = 48.2$, against a claimed mean $\mu_0 = 50$ with $\sigma = 6$. Test at the 5% level (two-tailed) whether the mean differs from 50, using critical values ± 1.96 .

Hypothesis test for a Poisson mean

- 9 A website historically crashes at a rate of $\lambda_0 = 4$ per month. After a software update, 1 crash is observed in a randomly selected month. Test at the 5% level whether the crash rate has decreased, using $P(X \leq 1) \approx 0.0916$ for $X \sim \text{Po}(4)$.
- 10 Explain why an exact Poisson probability, rather than a normal approximation, is used to test a hypothesis about a Poisson mean when λ_0 is small.
-

Further Poisson mean tests

- 11 A call center historically receives calls at $\lambda_0 = 6$ per hour. During a promotional event, 12 calls are received in one hour. Test at the 5% level whether the call rate has increased, using $P(X \geq 12) \approx 0.0201$ for $X \sim \text{Po}(6)$.
- 12 A factory's defect rate is historically $\lambda_0 = 3$ per shift. During a trial of new equipment, 8 defects are observed in one shift. Test at the 5% level whether the defect rate has increased, using $P(X \geq 8) \approx 0.0119$ for $X \sim \text{Po}(3)$.
-

Hypothesis test for a binomial proportion

- 13 A company claims 40% of its customers use its mobile app ($p_0 = 0.4$). In a sample of $n = 15$ customers, 10 report using the app. Test at the 5% level whether the true proportion is greater than 40%, using $P(X \geq 10) \approx 0.0338$ for $X \sim B(15, 0.4)$.
- 14 Explain why the binomial hypothesis test uses the exact binomial distribution rather than a normal approximation when n is small.
-

Further binomial proportion tests

- 15 A coin is tested for fairness ($p_0 = 0.5$). In 20 tosses, 6 heads are observed. Test at the 5% level (two-tailed) whether the coin is biased, using $P(X \leq 6) \approx 0.0577$ for $X \sim B(20, 0.5)$ (doubled for a two-tailed test).
- 16 A politician claims 60% of voters support a new policy ($p_0 = 0.6$). In a sample of 12 voters, only 3 support it. Test at the 5% level whether support is less than 60%, using $P(X \leq 3) \approx 0.0153$ for $X \sim B(12, 0.6)$.
-

Type I and Type II errors, and significance level

- 17 In the website crash-rate test, describe what a Type I error would mean in context, and what a Type II error would mean.

- 18 Explain what is meant by the significance level of a hypothesis test, and state the probability of a Type I error in a test conducted at the 5% significance level.
- 19 A researcher conducts a test at $\alpha = 0.10$ instead of the more common $\alpha = 0.05$. Explain how this affects the probability of Type I and Type II errors, compared to using $\alpha = 0.05$.
-

A well-designed word problem: testing a new manufacturing process

- 20 This question has four parts. A factory's current process produces components with mean breaking strength $\mu_0 = 250$ N and known $\sigma = 20$ N. A new process gives a sample of $n = 40$ components with mean $\bar{x} = 257$ N. (a) State H_0 and H_a for testing whether the new process increases mean breaking strength (one-tailed). (b) Compute the test statistic z . (c) Using critical value $z^* = 1.645$ (5% level), state the conclusion. (d) Quality control also monitors defects; the historical rate is $\lambda_0 = 5$ per batch, but a batch from the new process has only 1 defect. Using $P(X \leq 1) \approx 0.0404$ for $X \sim \text{Po}(5)$, test at the 5% level whether the new process has reduced the defect rate.